



NCI Protocol #: 10323

Patient ID:

Date:

1 of 22

Patient Name:
Date Collected:
Biopsy Site: **Liver**
Sample ID:
Telephone:
Tumor Content (%): **10_19%**

Patient DOB:
Referring Physician:
Primary Diagnosis: **Metastatic colorectal Cancer**
MoCha ID:
Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

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Report Highlights

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Relevant Colorectal Cancer Findings

Gene	Finding
BRAF	None detected
KRAS	KRAS G12D
NRAS	None detected
NTRK1	None detected
NTRK2	None detected
NTRK3	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	KRAS G12D Prognostic significance: None Diagnostic significance: None	None	None	46
IIC	ERBB2 V777L Prognostic significance: None Diagnostic significance: None	None	trastuzumab deruxtecan	21

Public data sources included in relevant therapies: FDA1, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.03(006). The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
III	FANCA N691S Prognostic significance: None Diagnostic significance: None	None	None	19
IID	TP53 D259Y Prognostic significance: None Diagnostic significance: None	None	None	15
IIC	ARID1A S1684fs Prognostic significance: None Diagnostic significance: None	None	None	3

Public data sources included in relevant therapies: FDA¹, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

 Alerts informed by public data sources:  Contraindicated,  Resistance

KRAS G12D cetuximab¹, panitumumab¹

Public data sources included in alerts: FDA¹, NCCN

Prevalent cancer biomarkers without relevant evidence based on included data sources
SMAD4 D355V

Variant Details

DNA Sequence Variants							
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Transcript	Variant Effect
ARID1A	p.(S1684fs)	c.5051_5052insAG	.	chr1:27102119	13.18%	NM_006015.4	frameshift Insertion
KRAS	p.(G12D)	c.35G>A	COSM521	chr12:25398284	14.07%	NM_033360.3	missense
FANCA	p.(N691S)	c.2072A>G	.	chr16:89838165	13.55%	NM_000135.2	missense
TP53	p.(D259Y)	c.775G>T	.	chr17:7577506	15.13%	NM_000546.5	missense
ERBB2	p.(V777L)	c.2329G>T	COSM14062	chr17:37881000	16.10%	NM_004448.3	missense
SMAD4	p.(D355V)	c.1064A>T	.	chr18:48591901	17.86%	NM_005359.5	missense

Comments:
N/A

Signed By: _____

Biomarker Descriptions

ARID1A (AT-rich interaction domain 1A)

Background: The ARID1A gene encodes the AT-rich interaction domain 1A tumor suppressor protein¹. ARID1A, also known as BAF250A, belongs to the ARID1 subfamily that also includes ARID1B^{1,2}. ARID1A and ARID1B are mutually exclusive subunits of the BAF variant of the SWI/SNF chromatin-remodeling complex^{2,3}. The BAF complex is a multisubunit protein that consists of SMARCB1/IN1, SMARCC1/BAF155, SMARCC2/BAF170, SMARCA4/BRG1, and SMARCA2/BRM³. The BAF complex remodels chromatin at promoter and enhancer elements to alter and regulate gene expression^{3,4}. ARID1A binds to transcription factors and coactivator/corepressor complexes to alter transcription². Recurrent inactivating mutations in BAF complex subunits, including ARID1A, lead to transcriptional dysfunction thereby, altering its tumor suppressor function².

Alterations and prevalence: Mutations in SWI/SNF complex subunits are the most commonly mutated chromatin modulators in cancer and have been observed in 20% of all tumors⁴. The majority of ARID1A inactivating mutations are nonsense or frameshift mutations². Somatic mutations in ARID1A have been identified in 50% of ovarian clear cell carcinoma, 30% of endometrioid carcinoma, and 24-43% of uterine corpus endometrial carcinoma, bladder urothelial carcinoma, and stomach adenocarcinoma^{3,5,6}. In microsatellite stable (MSS) colorectal cancer, mutations in ARID1A have been observed to correlate with increased tumor mutational burden (TMB) and expression of genes involved in the immune response⁷.

Potential relevance: Currently, no therapies are approved for ARID1A aberrations.

ERBB2 (erb-b2 receptor tyrosine kinase 2)

Background: The ERBB2 gene encodes the erb-b2 receptor tyrosine kinase 2, a member of the human epidermal growth factor receptor (HER) family. Along with ERBB2/HER2, EGFR/ERBB1/HER1, ERBB3/HER3, and ERBB4/HER4 make up the HER protein family⁸. All ERBB/HER proteins encode transmembrane receptor tyrosine kinases. However, ERBB2/HER2 is an orphan receptor with no known ligand. ERBB2 preferentially binds other ligand bound ERBB/HER family members to form hetero-dimers resulting in the activation of ERBB2 tyrosine kinase activity and subsequent activation of the PI3K/AKT/MTOR and RAS/RAF/MAPK/ERK signaling pathways which promote cell proliferation, differentiation, and survival⁹. Recurrent focal amplification of the ERBB2 gene leads to increased expression in several cancer types. ERBB2 overexpression in immortalized cell lines is oncogenic and leads to ERBB2 homo-dimerization and activation without ligand binding^{10,11,12}.

Alterations and prevalence: ERBB2 gene amplification occurs in 10-20% of breast, esophageal, and gastric cancers, 5-10% of bladder, cervical, pancreas, and uterine cancers, and 1-5% of colorectal, lung, and ovarian cancers^{5,6,13,14,15,16,17,18}. Recurrent somatic activating mutations in ERBB2/HER2 occur at low frequencies (<1%) in diverse cancer types^{6,19,20}. In breast, bladder, and colorectal cancers, the most common recurrent ERBB2 activating mutations include kinase domain mutations L755S and V777L and the extracellular domain mutation S310F. In lung cancer, the most common recurrent ERBB2 activating mutations include in-frame exon 20 insertions, particularly Y772_A775dup.

Potential relevance: The discovery of ERBB2/HER2 as an important driver of breast cancer in 1987 led to the development of trastuzumab, a humanized monoclonal antibody with specificity to the extracellular domain of HER2^{21,22}. Trastuzumab²³ was FDA approved for the treatment of HER2 positive breast cancer in 1998, and subsequently in HER2 positive metastatic gastric and gastroesophageal junction adenocarcinoma in 2010. Additional monoclonal antibody therapies have been approved by the FDA for HER2-positive breast cancer including pertuzumab²⁴ (2012), a humanized monoclonal antibody that inhibits HER2 dimerization, and ado-trastuzumab emtansine²⁵ (2013), a conjugate of trastuzumab and a potent antimicrotubule agent. The combination of pertuzumab, trastuzumab, and a taxane is the preferred front-line regimen for HER2-positive metastatic breast cancer²⁶. In addition to

Biomarker Descriptions (continued)

monoclonal antibodies, the small molecule inhibitor lapatinib²⁷, with specificity for both EGFR and ERBB2, was FDA approved (2007) for the treatment of patients with advanced HER2-positive breast cancer who have received prior therapy including trastuzumab. In 2017, the FDA approved the use of neratinib²⁸, an irreversible kinase inhibitor of EGFR, ERBB2/HER2, and ERBB4, for the extended adjuvant treatment of adult patients with early stage HER2-positive breast cancer. In 2020, the FDA approved neratinib²⁸ in combination with capecitabine for HER2-positive advanced or metastatic patients after two or more prior HER2-directed therapies. Also in 2020, the TKI irbinitinib²⁹ was FDA approved for HER2 overexpressing or amplified breast cancer in combination with trastuzumab and capecitabine. In 2021, the PD-1 blocking antibody, pembrolizumab, in combination with trastuzumab, fluoropyrimidine- and platinum-based chemotherapy, was approved for HER2 amplified gastric or gastroesophageal (GEJ) adenocarcinoma in the first line³⁰. The vaccine, nelipepimut-S³¹, was granted fast-track designation by the FDA (2016) in patients with low to intermediate HER2 expressing (IHC score 1+ or 2+) breast cancer. In 2018 fast-track designation was granted to the monoclonal antibody margetuximab³² in patients with ERBB2 positive breast cancer previously treated with an anti-HER2 therapy. In 2019, fast track designation was granted to the HER2-targeting antibody drug conjugate, amcenestrant³³, for HER2-positive advanced or metastatic breast cancer after one or more prior anti-HER2 based regimens. Additionally, in 2019, the novel bispecific antibody, zanidatamab³⁴, received fast-track designation in combination with standard chemotherapy for patients with HER2-overexpressing gastroesophageal adenocarcinoma (GEA) and breakthrough therapy designation (2020) as a monotherapy for patients with HER2-amplified biliary tract cancer³⁵. In 2020, BDTX-189³⁶ received fast-track designation for adult patients with solid tumors harboring an allosteric human ERBB2 mutation or exon 20 insertion, and the humanized anti-HER2 antibody drug conjugate disitamab vedotin received breakthrough designation for adult patients with HER2-positive urothelial cancer after previous platinum-chemotherapy treatment³⁷. In 2021, the antibody-drug conjugate ARX788³⁸ received fast-track designation as a monotherapy for advanced or metastatic HER2-positive breast cancer that have progressed on one or more anti-HER2 regimens. Additionally, in 2021, fast track designation was granted to HER2 targeted chimeric antigen receptor macrophage (CAR-M), CT-0508³⁹, for HER2-overexpressing solid tumors. Certain activating mutations have been observed to impart sensitivity to neratinib, afatinib, lapatinib, and trastuzumab, or dacomitinib in early and ongoing clinical studies^{40,41,42,43,44}. ERBB2 kinase domain mutations R896G and V659E both showed response to afatinib in two NSCLC case studies^{45,46}. Additionally, acquired HER2 mutations in estrogen receptor-positive (ER+) breast cancer have been shown to confer resistance to hormone therapy⁴⁷. However, this was shown to be overcome by neratinib in combination with therapies targeting ER⁴⁷.

FANCA (FA complementation group A)

Background: The FANCA gene encodes the FA complementation group A protein, a member of the Fanconi Anemia (FA) family, which also includes FANCB, FANCC, FANCD1 (BRCA2), FANCD2, FANCE, FANCF, FANCG, FANCI, FANCL, FANCM, and FANCN (PALB2)¹. FA genes are tumor suppressors that are responsible for the maintenance of replication fork stability, DNA damage repair through the removal of interstrand cross-links (ICL), and subsequent initiation of the homologous recombination repair (HRR) pathway^{48,49}. In response to DNA damage, FANCA, FANCB, FANCC, FANCE, FANCF, FANCG, FANCL, and FANCM assemble to form the FA core complex which is responsible for the monoubiquitination of the FANCI-FANCD2 (ID2) complex⁴⁸. Monoubiquitination of the ID2 complex promotes co-localization with BRCA1/2, which is critical in BRCA mediated DNA repair^{50,51}. Loss of function mutations in the FA family and HRR pathway, including FANCA, can result in the BRCAness phenotype, characterized by a defect in the HRR pathway, mimicking BRCA1 or BRCA2 loss^{52,53}. Germline mutations in FA genes lead to Fanconi Anemia, a condition characterized by chromosomal instability and congenital abnormalities, including bone marrow failure and cancer predisposition^{54,55}. Of those diagnosed with FA, mutations in FANCA are the most common and confer predisposition to myelodysplastic syndrome, acute myeloid leukemia, and solid tumors^{49,55,56,57,58}.

Alterations and prevalence: Somatic mutations in FANCA are observed in 4-8% of uterine, colorectal, and bladder cancers and about 6% of melanoma⁵. Biallelic loss is also reported in 2-5% of uveal melanoma, invasive breast carcinoma, ovarian cancer, and prostate cancer⁵.

Potential relevance: Currently, no therapies are approved for FANCA aberrations. Consistent with other genes that contribute to the BRCAness phenotype, mutations in FANCA are shown to confer enhanced sensitivity in vitro to DNA damaging agents, including cisplatin, as well as PARP inhibitors such as olaparib^{59,60}. FANCA copy number loss along with reduced expression has also been associated with genetic instability in sporadic acute myeloid leukemia (AML)⁵⁸.

KRAS (KRAS proto-oncogene, GTPase)

Background: The KRAS proto-oncogene encodes a GTPase that functions in signal transduction and is a member of the RAS superfamily which also includes NRAS and HRAS. RAS proteins mediate the transmission of growth signals from the cell surface to the nucleus via the PI3K/AKT/MTOR and RAS/RAF/MEK/ERK pathways, which regulate cell division, differentiation, and survival^{61,62,63}.

Biomarker Descriptions (continued)

Alterations and prevalence: Recurrent mutations in RAS oncogenes cause constitutive activation and are found in 20-30% of cancers. KRAS mutations are observed in up to 10-20% of uterine cancer, 30-35% of lung adenocarcinoma and colorectal cancer, and about 60% of pancreatic cancer⁵. The majority of KRAS mutations consist of point mutations occurring at G12, G13, and Q61^{5,64,65}. Mutations at A59, K117, and A146 have also been observed but are less frequent^{6,66}.

Potential relevance: The KRAS inhibitor, sotorasib⁶⁷, is approved (2021) for the treatment of adult patients with KRAS G12C-mutated locally advanced or metastatic non-small cell lung cancer (NSCLC). The FDA has granted breakthrough therapy designation (2021) to the small molecule inhibitor, adagrasib, for KRAS G12C positive in non-small cell lung cancer following prior systemic therapy⁶⁸. The small molecular inhibitor, RO-5126766, was also granted breakthrough designation (2021) alone for KRAS G12V mutant non-small cell lung cancer or in combination with defactinib, for KRAS mutant endometrial carcinoma and KRAS G12V mutant non-small cell lung cancer⁶⁹. Additionally, onvansertib⁷⁰ was granted fast track designation (2020) for second-line treatment of patients with KRAS-mutated metastatic colorectal cancer (mCRC). The EGFR antagonists, cetuximab⁷¹ and panitumumab⁷², are contraindicated for treatment of colorectal cancer patients with KRAS mutations in exon 2 (codons 12 and 13), exon 3 (codons 59 and 61), and exon 4 (codons 117 and 146)⁶⁶. Additionally, KRAS mutations are associated with poor prognosis in NSCLC⁷³.

SMAD4 (SMAD family member 4)

Background: The SMAD4 gene encodes the SMAD family member 4, a transcription factor that belongs to a family of 8 SMAD genes that can be divided into three main classes. SMAD4 (also known as DPC4) belongs to the common mediator SMAD (co-SMAD) class while SMAD1, SMAD2, SMAD3, SMAD5, and SMAD8 are part of the regulator SMAD (R-SMAD) class. The inhibitory SMAD (I-SMAD) class includes both SMAD6 and SMAD7^{74,75}. SMAD4 is a tumor suppressor gene and functions as a mediator of the TGF- β and BMP signaling pathways that are implicated in cancer initiation and progression^{75,76,77}. Loss of SMAD4 does not drive oncogenesis, but is associated with progression of cancers initiated by driver genes such as KRAS and APC^{74,75}.

Alterations and prevalence: Inactivation of SMAD4 can occur due to mutations, allelic loss, homozygous deletions, and 18q loss of heterozygosity (LOH)⁷⁴. Somatic mutations in SMAD4 occur in up to 20% of pancreatic, 12% of colorectal, and 8% of stomach cancers. Recurrent hotspot mutations including R361 and P356 occur in the mad homology 2 (MH2) domain leading to the disruption of the TGF- β signaling^{6,77,78}. Copy number deletions occur in up to 12% of pancreatic, 10% of esophageal, and 13% of stomach cancers^{5,6,14}.

Potential relevance: Currently, no therapies are approved for SMAD4 aberrations. Clinical studies and meta-analyses have demonstrated that loss of SMAD4 expression confers poor prognosis and poor overall survival (OS) in colorectal and pancreatic cancers^{75,77,79,80,81}. Importantly, SMAD4 is a predictive biomarker to fluorouracil based chemotherapy^{82,83}. In a retrospective analysis of 241 colorectal cancer patients treated with fluorouracil, 21 patients with SMAD4 loss demonstrated significantly poor median OS when compared to SMAD4 positive patients (31 months vs 89 months)⁸³. In another clinical study of 173 newly diagnosed and recurrent head and neck squamous cell carcinoma (HNSCC) patients, SMAD4 loss is correlated with cetuximab resistance in HPV-negative HNSCC tumors⁸⁴.

TP53 (tumor protein p53)

Background: The TP53 gene encodes the p53 tumor suppressor protein that binds to DNA and activates transcription in response to diverse cellular stresses to induce cell cycle arrest, apoptosis, or DNA repair. In unstressed cells, TP53 is kept inactive by targeted degradation via MDM2, a substrate recognition factor for ubiquitin-dependent proteolysis. Alterations in TP53 is required for oncogenesis as they result in loss of protein function and gain of transforming potential⁸⁵. Germline mutations in TP53 are the underlying cause of Li-Fraumeni syndrome, a complex hereditary cancer predisposition disorder associated with early-onset cancers^{86,87}.

Alterations and prevalence: TP53 is the most frequently mutated gene in the cancer genome with approximately half of all cancers experiencing TP53 mutations. Ovarian, head and neck, esophageal, and lung squamous cancers have particularly high TP53 mutation rates (60-90%)^{5,6,88,89,90,91}. Approximately two-thirds of TP53 mutations are missense mutations and several recurrent missense mutations are common including substitutions at codons R158, R175, Y220, R248, R273, and R282^{5,6}. Invariably, recurrent missense mutations in TP53 inactivate its ability to bind DNA and activate transcription of target genes^{92,93,94,95}.

Potential relevance: The small molecule p53 reactivator, PC14586, received a fast track designation (2020) by the FDA for advanced tumors harboring a TP53 Y220C mutation⁹⁶. The FDA has granted fast track designation (2019) to the p53 reactivator, eprenetapopt⁹⁷ and breakthrough designation⁹⁸ (2020) in combination with azacitidine or azacitidine and venetoclax for acute myeloid leukemia patients (AML) and myelodysplastic syndrome (MDS) harboring a TP53 mutation, respectively. In addition to investigational therapies

Biomarker Descriptions (continued)

aimed at restoring wild-type TP53 activity, compounds that induce synthetic lethality are also under clinical evaluation^{99,100}. TP53 mutations confer poor prognosis in multiple blood cancers including AML, MDS, myeloproliferative neoplasms (MPN), and chronic lymphocytic leukemia (CLL)^{101,102,103,104}. In mantle cell lymphoma, TP53 mutations are associated with poor prognosis when treated with conventional therapy including hematopoietic cell transplant¹⁰⁵. Mono- and bi-allelic mutations in TP53 confer unique characteristics in MDS, with multi-hit patients also experiencing associations with complex karyotype, few co-occurring mutations, and high-risk disease presentation as well as predicted death and leukemic transformation independent of the IPSS-R staging system¹⁰⁶.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KRAS G12D

Relevant Therapy	FDA	NCCN	Clinical Trials*
apatinib, chemotherapy	×	×	● (IV)
bevacizumab, chemotherapy	×	×	● (III)
atezolizumab + cobimetinib	×	×	● (II)
cabozantinib, nivolumab	×	×	● (II)
cetuximab, chemotherapy	×	×	● (II)
nivolumab, chemotherapy, bevacizumab	×	×	● (II)
panitumumab, trametinib	×	×	● (II)
pembrolizumab	×	×	● (II)
selinexor, pembrolizumab	×	×	● (II)
ASN007, palbociclib	×	×	● (I/II)
ASTX029	×	×	● (I/II)
cobimetinib, atezolizumab, antimalarial	×	×	● (I/II)
eltanexor	×	×	● (I/II)
HH-2710	×	×	● (I/II)
mirdametinib, lifirafenib	×	×	● (I/II)
navitoclax, trametinib	×	×	● (I/II)
neratinib, valproic acid	×	×	● (I/II)
OMO-103	×	×	● (I/II)
onvansertib, chemotherapy, bevacizumab	×	×	● (I/II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

KRAS G12D (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
regorafenib, chemotherapy	×	×	● (I/II)
repotrectinib, trametinib	×	×	● (I/II)
RMC-4630, pembrolizumab	×	×	● (I/II)
SX-682, nivolumab	×	×	● (I/II)
zotatifin	×	×	● (I/II)
AZD-0364	×	×	● (I)
BBP-398	×	×	● (I)
BI-1701963, BI-3011441	×	×	● (I)
BI-3011441	×	×	● (I)
CDK 4/6 inhibitor, MEK inhibitor	×	×	● (I)
chemotherapy, binimetinib	×	×	● (I)
cobimetinib, belvarafenib	×	×	● (I)
COM701, nivolumab	×	×	● (I)
DCC-3116, trametinib	×	×	● (I)
GH35	×	×	● (I)
HBI 2376	×	×	● (I)
JAB-3312	×	×	● (I)
JSI-1187	×	×	● (I)
neratinib, trametinib	×	×	● (I)
palbociclib, binimetinib	×	×	● (I)
PF-07284892, binimetinib	×	×	● (I)
RMC-4630	×	×	● (I)
TAS-0612	×	×	● (I)
TNO-155, ribociclib	×	×	● (I)
trametinib, ruxolitinib	×	×	● (I)
ulixertinib, antimalarial	×	×	● (I)

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Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

ERBB2 V777L

Relevant Therapy	FDA	NCCN	Clinical Trials*
trastuzumab deruxtecan	×	○	● (II)
pyrotinib	×	×	● (IV)
ado-trastuzumab emtansine	×	×	● (II)
ado-trastuzumab emtansine + atezolizumab, pertuzumab/trastuzumab/hyaluronidase-zzxf, pertuzumab/trastuzumab/hyaluronidase-zzxf + chemotherapy, ado-trastuzumab emtansine + irbinitinib	×	×	● (II)
afatinib	×	×	● (II)
atezolizumab + pertuzumab + trastuzumab	×	×	● (II)
irbinitinib, trastuzumab	×	×	● (II)
neratinib	×	×	● (II)
pertuzumab + trastuzumab	×	×	● (II)
pertuzumab + trastuzumab, irbinitinib + trastuzumab	×	×	● (II)
poziotinib	×	×	● (II)
trastuzumab, pertuzumab, ado-trastuzumab emtansine, lapatinib	×	×	● (II)
zotatifin	×	×	● (I/II)
ARX-788	×	×	● (I)
KN026, KN046	×	×	● (I)
neratinib, palbociclib, everolimus, trametinib	×	×	● (I)
SAR-443216	×	×	● (I)
SHR-A1811	×	×	● (I)
TAS 2940	×	×	● (I)

FANCA N691S

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab	×	×	● (II)
atezolizumab + talazoparib	×	×	● (II)
olaparib	×	×	● (II)

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Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

FANCA N691S (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
olaparib, pembrolizumab	×	×	● (II)
rucaparib	×	×	● (II)
rucaparib, atezolizumab	×	×	● (II)
talazoparib	×	×	● (II)
ATRN-119	×	×	● (I/II)
BAY-1895344	×	×	● (I/II)
venadaparib	×	×	● (I/II)
BAY-1895344, niraparib	×	×	● (I)
BAY-1895344, pembrolizumab	×	×	● (I)
berzosertib, chemotherapy	×	×	● (I)
copanlisib, olaparib, durvalumab	×	×	● (I)
HWH-340	×	×	● (I)
RP12146	×	×	● (I)
talazoparib, palbociclib, axitinib, crizotinib	×	×	● (I)

TP53 D259Y

Relevant Therapy	FDA	NCCN	Clinical Trials*
atezolizumab	×	×	● (II)
olaparib	×	×	● (II)
olaparib, pembrolizumab	×	×	● (II)
talazoparib	×	×	● (II)
ATRN-119	×	×	● (I/II)
venadaparib	×	×	● (I/II)
AMG-650	×	×	● (I)
BAY-1895344, niraparib	×	×	● (I)
BAY-1895344, pembrolizumab	×	×	● (I)

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Relevant Therapy Summary (continued)

In this cancer type

In other cancer type

In this cancer type and other cancer types

No evidence

TP53 D259Y (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
berzosertib, chemotherapy	×	×	● (I)
HWH-340	×	×	● (I)
LY3143921	×	×	● (I)
RP12146	×	×	● (I)
talazoparib, palbociclib, axitinib, crizotinib	×	×	● (I)

ARID1A S1684fs

Relevant Therapy	FDA	NCCN	Clinical Trials*
talazoparib	×	×	● (II)
CCS-1477	×	×	● (I/II)
copanlisib, olaparib, durvalumab	×	×	● (I)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

KRAS G12D

NCT ID	Title	Phase
NCT04835714	A Phase I Open-Label Dose Escalation Trial Of BI 1701963 As Monotherapy And In Combination With BI 3011441 In Patients With KRAS Mutated Advanced Or Metastatic Solid Tumours	I
NCT04045496	A Phase I, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of JAB-3312 in Adult Patients With Advanced Solid Tumors	I
NCT04000529	A Phase Ib, Open-label, Multi-center Study to Characterize the Safety, Tolerability, and Preliminary Efficacy of TNO155 in Combination With Spartalizumab or Ribociclib in Selected Malignancies	I
NCT02885753	Systemic Oxaliplatin or Intra-arterial Chemotherapy Combined With LV5FU2 and an Target Therapy in First Line Treatment of Metastatic Colorectal Cancer Restricted to the Liver	III
NCT03087071	A Phase II Enrichment Study of Panitumumab as a Single Agent or in Combination With Trametinib in Anti-EGFR-Refractory Stage IV Colorectal Cancer Patients	II
NCT04214418	Phase I/II Open-label Study of Combination Therapy With The MEK Inhibitor, Cobimetinib, Immune Checkpoint Blockade, Atezolizumab, And The AUTOPhagy Inhibitor, Hydroxychloroquine In KRAS-mutated Advanced Malignancies	I/II

Clinical Trials Summary (continued)

KRAS G12D (continued)

NCT ID	Title	Phase
NCT03829410	A Phase Ib/II Study of Onvansertib (PCM-075) in Combination With FOLFIRI and Bevacizumab for Second Line Treatment of Metastatic Colorectal Cancer in Patients With a KRAS Mutation.	I/II
No NCT ID	Exploratory Study Of Apatinib For Advanced Colorectal Cancer	IV
NCT04189055	Cetuximab as Salvage Therapy in Patients with Neo Wild-type RAS/RAF Metastatic Colorectal Cancer with Liver Metastases. A Proof-of-concept Study	II
NCT03519412	Pembrolizumab in MMR-Proficient Metastatic Colorectal Cancer Pharmacologically Primed to Trigger Dynamic Hypermutation Status	II
NCT05039177	A Phase 1b/2 Study of Agents Targeting the Mitogen-Activated Protein Kinase Pathway in Patients With Advanced Gastrointestinal Malignancies (HERKULES-3)	I/II
NCT04418661	A Phase 1/2, Open-label, Multicenter, Dose Escalation and Dose Expansion Study of SAR442720 in Combination With Pembrolizumab in Patients With Advanced Malignancies	I/II
NCT04615312	Phase I Clinical Study on the Safety and Tolerability of a CDK4 / 6 Inhibitor and a MEK Inhibitor in the Treatment of Metastatic Digestive System Tumors	I
NCT02613650	A Phase Ib Trial of a Combination of mFOLFIRI With MEK162 in Patients With Advanced RAS (HRAS, NRAS, or KRAS) Positive Metastatic Colorectal Cancers	I
NCT03667716	A Phase Ia/Ib Study of COM701 as Monotherapy and In Combination With an Anti-PD-1 Antibody in Subjects With Advanced Solid Tumors	I
NCT04892017	A Phase I, First-in-Human Study of DCC-3116 as a Single Agent and in Combination With Trametinib in Patients With Advanced or Metastatic Solid Tumors With RAS or RAF Mutations.	I
NCT04870034	Perioperative Analysis of Binimetinib and Palbociclib in RAS-Driven Tumors	I
NCT04145297	A Phase I Trial of Ulixertinib (BVD-523) and Hydroxychloroquine in Patients With Advanced MAPK-Mutated Gastrointestinal Adenocarcinomas	I
NCT02649790	A Phase I/II Open-Label Study of the Safety, Tolerability and Efficacy of the Selective Inhibitor of Nuclear Export (SINE) Compound Eltanexor (KPT-8602) in Patients With Newly Diagnosed and Relapsed/Refractory Cancer Indications	I/II
NCT04781270	mFOLFOXIRI Plus Bevacizumab Versus mFOLFOX6 Plus Bevacizumab for the First Line Treatment of RAS Mutant Unresectable Colorectal Liver-limited Metastases	III
NCT04072198	Phase II Study on NIVolumab in Combination With FOLFOXIRI/Bevacizumab in First Line Chemotherapy of Advanced COloRectal Cancer RASm/BRAFm Patients.	II
NCT04854434	A Phase II Open-Label Multicenter Study to Evaluate the Safety and Efficacy of Selinexor With or Without Pembrolizumab Versus Standard of Care in Previously Treated Metastatic Colorectal Cancer With RAS Mutations	II
NCT03905148	A Phase Ib, Open-Label, Dose-escalation and Expansion Study to Investigate the Safety, Pharmacokinetics and Antitumor Activities of a RAF Dimer Inhibitor BGB-283 in Combination With MEK Inhibitor PD-0325901 in Patients With Advanced or Refractory Solid Tumors	I/II
NCT04808362	A Phase I/II Study to Evaluate the Safety, Pharmacokinetics, and Anti-Tumour Activity of the MYC Inhibitor OMO-103 Administered Intravenously in Patients With Advanced Solid Tumours	I/II

Clinical Trials Summary (continued)

KRAS G12D (continued)

NCT ID	Title	Phase
NCT03828799	Safety, Tolerability and Efficacy of Regorafenib in Combination with FOLFIRINOX in Patients with RAS-mutated Metastatic Colorectal Cancer: a Dose-escalation, Phase I/II Trial	I/II
NCT03284502	A Phase Ib, Open-label, Multicenter, Dose Escalation Study of the Safety, Tolerability, and Pharmacokinetics of HM95573 in Combination With Either Cobimetinib or Cetuximab in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04963283	A Phase II Study of Cabozantinib and Nivolumab in Refractory Metastatic Microsatellite Stable (MSS) Colorectal Cancer	II
NCT05071183	A Phase I/II Study of Repotrectinib in Combination with Trametinib in Patients with KRAS Mutant Non-Small Cell Lung Cancer	I/II
NCT04586270	A Phase I Study of TAS0612 in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT02079740	An Open Label, Two-Part, Phase Ib/II Study to Investigate the Safety, Pharmacokinetics, Pharmacodynamics, and Clinical Activity of the MEK Inhibitor Trametinib and the BCL2-Family Inhibitor Navitoclax (ABT-263) in Combination in Subjects With KRAS or NRAS Mutation-Positive Advanced Solid Tumors	I/II
NCT04092673	A Phase 1-2 Dose-Escalation and Cohort-Expansion Study of Intravenous Zotatfin (eFT226) in Subjects With Selected Advanced Solid Tumor Malignancies	I/II
NCT04742556	A Phase I Open-label Trial of BI 3011441 in Japanese Patients With NRAS/KRAS Mutation Positive Advanced, Unresectable or Metastatic Refractory Solid Tumours	I
NCT03919292	Phase 1/2 Study of Neratinib and Divalproex Sodium (Valproate) in Advanced Solid Tumors, With an Expansion Cohort in Ras-Mutated Cancers	I/II
NCT04599140	Phase Ib/II Trial of SX-682 in Combination With Nivolumab for Refractory RAS Mutated (RAS) Microsatellite Stable (MSS) Metastatic Colorectal Cancer (mCRC) (STOPTRAFFIC-1)	I/II
NCT05010694	A Phase I Study Evaluating the Safety, Tolerability, Pharmacokinetic Characteristics, and Primary Antitumor Activity of GH35 in Patients With Advanced Solid Tumors With KRAS Mutation	I
NCT05163028	A Phase I, Open-Label, Dose Escalation of HBI-2376 in Patients With Advanced Malignant Solid Tumors Harboring KRAS or EGFR Mutations	I
NCT04418167	A Phase I Study of ERK1/2 Inhibitor JSI-1187 Administered as Monotherapy and in Combination With Dabrafenib for the Treatment of Advanced Solid Tumors With MAPK Pathway Mutations	I
NCT03065387	Phase I Study of the Pan-ERBB Inhibitor Neratinib Given in Combination With Everolimus, Palbociclib, or Trametinib in Advanced Cancer Subjects With EGFR Mutation/Amplification, HER2 Mutation/Amplification, or HER3/4 Mutation or KRAS Mutation	I
NCT04800822	A Phase I Open-Label, Multi-Center, Dose Escalation and Dose Expansion Study to Evaluate the Safety, Tolerability, Pharmacokinetics, and Preliminary Evidence of Antitumor Activity of PF-07284892 (ARRY-558) as a Single Agent and in Combination Therapy in Participants With Advanced Solid Tumors	I
NCT04303403	Phase Ib Study Evaluating Safety and Tolerability of Combination Trametinib and Ruxolitinib in Patients with Advanced RAS Mutant Colorectal Cancer and Pancreatic Adenocarcinoma	I
NCT04198818	A First-in-Human, Open Label, Phase I/II Study to Evaluate the Safety, Tolerability and Pharmacokinetics of HH2710 in Patients With Advanced Tumors	I/II

Clinical Trials Summary (continued)

KRAS G12D (continued)

NCT ID	Title	Phase
NCT03634982	A Phase I, Open-Label, Multicenter, Dose-Escalation Study of RMC-4630 Monotherapy in Adult Participants with Relapsed/Refractory Solid Tumors	I
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT03520075	A Phase I/II Study of the Safety, Pharmacokinetics, and Activity of ASTX029 in Subjects With Advanced Solid Tumors	I/II
NCT04305249	A Phase I, Open-Label, Multi-Center Dose Finding Study to Investigate the Safety, Pharmacokinetics, and Preliminary Efficacy of ATG-017 Monotherapy in Patients With Advanced Solid Tumors and Hematological Malignancies	I
NCT04528836	A Phase I/IB First-in-Human Study of the SHP2 Inhibitor BBP-398 (Formerly Known as IACS-15509) in Patients With Advanced Solid Tumors	I

ERBB2 V777L

NCT ID	Title	Phase
NCT04172597	A Phase II Study of Pozitotinib in Patients With EGFR or HER2 Activating Mutations in Advanced Malignancies	II
NCT04591431	The Rome Trial From Histology to Target: the Road to Personalize Target Therapy and Immunotherapy	II
NCT04040699	A Phase Ib Study to Evaluate Efficacy, Safety and Tolerability of KN026 Combined With KN046 in Subjects With HER2 Positive Solid Tumor.	I
No NCT ID	Single Arm, Open Label, Signal Seeking, Phase IIa Trial Of The Activity Of Trastuzumab Emtansine (T-DM1) In Patients With Tumours Harboursing HER2 Amplifications Or Mutations	II
NCT04551521	Continuous ReAssessment With Flexible ExTension in Rare Malignancies - CRAFT: The NCT-PMO-1602 Phase II Trial	II
NCT04579380	A Phase II Basket Study of Tucatinib in Combination With Trastuzumab in Subjects With Previously Treated, Locally Advanced Unresectable or Metastatic Solid Tumors Driven by HER2 Alterations	II
NCT01953926	An Open-Label, Phase II Basket Study of Neratinib in Patients With Solid Tumors With Somatic Activating HER Mutations	II
NCT04639219	A Phase II, Multicenter, Open-label Study to Evaluate the Efficacy and Safety of Trastuzumab Deruxtecan (T-DXd) for the Treatment of Unresectable and/or Metastatic Solid Tumors Harboring HER2 Activating Mutations Regardless of Tumor Histology	II
NCT04092673	A Phase 1-2 Dose-Escalation and Cohort-Expansion Study of Intravenous Zotatfin (eFT226) in Subjects With Selected Advanced Solid Tumor Malignancies	I/II
NCT03255070	A Phase I, Multicenter, Open-label, Multiple Dose-escalation and Expansion Study of ARX788, as Monotherapy in Advanced Solid Tumors With HER2 Expression	I
No NCT ID	An Exploratory Basket Study of Non Breast Cancer Patients with HER2 Mutation or Amplification who Failed in Standard Treatment with Pyrotinib Maleate Tablets	IV

Clinical Trials Summary (continued)

ERBB2 V777L (continued)

NCT ID	Title	Phase
NCT04589845	Tumor-Agnostic Precision Immunooncology and Somatic Targeting Rational for You (TAPISTRY) Phase II Platform Trial	II
NCT04632992	MyTACTIC: An Open-Label Phase II Study Evaluating Targeted Therapies in Patients Who Have Advanced Solid Tumors With Genomic Alterations or Protein Expression Patterns Predictive of Response	II
NCT03810872	An Open Explorative Phase II, Open Label Study of Afatinib in the Treatment of Advanced Cancer Carrying an EGFR, a HER2 or a HER3 Mutation	II
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT04423185	Platform Study of Genotyping Guided Precision Medicine for Rare Tumors in China	II
NCT03065387	Phase I Study of the Pan-ERBB Inhibitor Neratinib Given in Combination With Everolimus, Palbociclib, or Trametinib in Advanced Cancer Subjects With EGFR Mutation/Amplification, HER2 Mutation/Amplification, or HER3/4 Mutation or KRAS Mutation	I
NCT04446260	A Phase I Multi-Country, Multi-Center, Open-Label Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Efficacy of SHR-A1811 in HER2 Expressing or Mutated Advanced Malignant Solid Tumor Subjects	I
NCT03297606	Canadian Profiling and Targeted Agent Utilization Trial (CAPTUR): A Phase II Basket Trial	II
NCT05013554	A Phase I/Ib Open-label, First-in-human, Single Agent, Dose Escalation and Expansion Study for the Evaluation of Safety, Pharmacokinetics, Pharmacodynamics, and Anti-tumor Activity of SAR443216 in Participants with Relapsed/Refractory HER2 Expressing Solid Tumors.	I
NCT04982926	A Phase I Study of TAS2940 in Patients With Locally Advanced or Metastatic Solid Tumors With EGFR and / or HER2 Aberrations	I

FANCA N691S

NCT ID	Title	Phase
NCT04095273	A Multicenter, Non-randomized, Open-label Phase Ib Study to Determine the Maximum Tolerated and Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Combination With Pembrolizumab and to Characterize Its Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumor Activity in Patients With Advanced Solid Tumors	I
NCT05002868	A Multi-center, Open-label, Phase I/Ib Study to Assess the Safety, Pharmacokinetics and Anti-tumor Activity of RP12146, a Poly (ADP-ribose) Polymerase (PARP) Inhibitor, in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT02693535	Targeted Agent and Profiling Utilization Registry (TAPUR) Study.	II
NCT04171700	A Phase II Multicenter, Open-label Study of Rucaparib as Treatment for Solid Tumors Associated With Deleterious Mutations in Homologous Recombination Repair Genes	II
NCT04276376	A Multicenter, Open Label, Phase II Basket Trial Exploring the Efficacy And Safety of The Combination of Rucaparib (PARP Inhibitor) And Atezolizumab (Anti-PD-L1 Antibody) In Patients With DNA Repair-Deficient or Platinum-Sensitive Solid Tumors	II

Clinical Trials Summary (continued)

FANCA N691S (continued)

NCT ID	Title	Phase
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT05071209	A Phase I/II Study of BAY 1895344 (Elimusertib, NSC#810486) in Pediatric Patients With Relapsed or Refractory Solid Tumors	I/II
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and Durvalumab (MEDI4736) in Patients With Advanced Solid Tumors	I
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients with Progressive Locally-advanced or Metastatic Solid Tumors.	II
NCT04123366	A Phase II Study of Olaparib in Combination With Pembrolizumab in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) and/or Homologous Recombination Deficiency (HRD)-Positive Advanced Cancer	II
NCT04174716	An Open-label, Multi-center, Phase Ib/Ila Basket Trial of IDX-1197 in Patients With Homologous Recombination Repair Mutated Solid Tumors	I/II
NCT03415659	A Phase I, Open-label, Single-center, Single/Multiple-dose, Dose-escalation/Dose-expansion Clinical Study on Tolerance and Pharmacokinetics of HWH340 Tablet in Patients With Advanced Solid Tumors	I
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT04905914	A Phase I/IIa, Open-Label, Safety, Pharmacokinetic, And Preliminary Efficacy Study Of Oral ATRN-119 In Patients With Advanced Solid Tumors	I/II
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT04267939	An Open-label Phase Ib Study to Determine the Maximum Tolerated and/or Recommended Phase II Dose of the ATR Inhibitor BAY 1895344 in Combination With PARP Inhibitor Niraparib, in Patients With Recurrent Advanced Solid Tumors and Ovarian Cancer	I
NCT02595931	Phase I Clinical Trial of M6620 (VX-970, Berzosertib) in Combination With the Topoisomerase I Inhibitor Irinotecan in Patients With Advanced Solid Tumors	I
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I

TP53 D259Y

NCT ID	Title	Phase
NCT03096054	A Cancer Research UK (CR-UK) Phase I Trial of LY3143921 a Cdc7 Inhibitor in Adult Patients With Advanced Solid Tumours	I

Clinical Trials Summary (continued)

TP53 D259Y (continued)

NCT ID	Title	Phase
NCT04095273	A Multicenter, Non-randomized, Open-label Phase Ib Study to Determine the Maximum Tolerated and Recommended Phase II Dose of the ATR Inhibitor BAY1895344 in Combination With Pembrolizumab and to Characterize Its Safety, Tolerability, Pharmacokinetics and Preliminary Anti-tumor Activity in Patients With Advanced Solid Tumors	I
NCT05002868	A Multi-center, Open-label, Phase I/Ib Study to Assess the Safety, Pharmacokinetics and Anti-tumor Activity of RP12146, a Poly (ADP-ribose) Polymerase (PARP) Inhibitor, in Patients With Locally Advanced or Metastatic Solid Tumors	I
NCT04293094	A Phase I, Multicenter, Open-label, Dose-Exploration and Dose-Expansion Study Evaluating the Safety, Tolerability, Pharmacokinetics, and Efficacy of AMG 650 in Subjects With Advanced Solid Tumors	I
NCT02029001	A Two-period, Multicenter, Randomized, Open-label, Phase II Study Evaluating the Clinical Benefit of a Maintenance Treatment Targeting Tumor Molecular Alterations in Patients with Progressive Locally-advanced or Metastatic Solid Tumors.	II
NCT04123366	A Phase II Study of Olaparib in Combination With Pembrolizumab in Participants With Previously Treated, Homologous Recombination Repair Mutation (HRRm) and/or Homologous Recombination Deficiency (HRD)-Positive Advanced Cancer	II
NCT02401347	A Phase II Clinical Trial of the PARP Inhibitor Talazoparib in BRCA1 and BRCA2 Wild Type Patients With Advanced Triple Negative Breast Cancer and Homologous Recombination Deficiency or Advanced HER2 Negative Breast Cancer or Other Solid Tumors With a Mutation in Homologous Recombination Pathway Genes Talazoparib Beyond BRCA (TBB) Trial	II
NCT04174716	An Open-label, Multi-center, Phase Ib/Ila Basket Trial of IDX-1197 in Patients With Homologous Recombination Repair Mutated Solid Tumors	I/II
NCT03415659	A Phase I, Open-label, Single-center, Single/Multiple-dose, Dose-escalation/Dose-expansion Clinical Study on Tolerance and Pharmacokinetics of HWH340 Tablet in Patients With Advanced Solid Tumors	I
NCT03767075	Basket of Baskets: A Modular, Open-label, Phase II, Multicentre Study To Evaluate Targeted Agents in Molecularly Selected Populations With Advanced Solid Tumours	II
NCT04905914	A Phase I/IIa, Open-Label, Safety, Pharmacokinetic, And Preliminary Efficacy Study Of Oral ATRN-119 In Patients With Advanced Solid Tumors	I/II
NCT03233204	NCI-COG Pediatric MATCH (Molecular Analysis for Therapy Choice)- Phase 2 Subprotocol of Olaparib in Patients With Tumors Harboring Defects in DNA Damage Repair Genes	II
NCT04267939	An Open-label Phase Ib Study to Determine the Maximum Tolerated and/or Recommended Phase II Dose of the ATR Inhibitor BAY 1895344 in Combination With PARP Inhibitor Niraparib, in Patients With Recurrent Advanced Solid Tumors and Ovarian Cancer	I
NCT02595931	Phase I Clinical Trial of M6620 (VX-970, Berzosertib) in Combination With the Topoisomerase I Inhibitor Irinotecan in Patients With Advanced Solid Tumors	I
NCT04693468	Modular Phase IB Hypothesis-Testing, Biomarker-Driven, Talazoparib Combination Trial (TalaCom)	I

Clinical Trials Summary (continued)

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NCT ID	Title	Phase
NCT04550494	A Pharmacodynamics-Driven Trial of Talazoparib, an Oral PARP Inhibitor, in Patients With Advanced Solid Tumors and Aberrations in Genes Involved in DNA Damage Response	II
NCT03568656	An Open-label Phase I/IIa Study to Evaluate the Safety and Efficacy of CCS1477 as Monotherapy and in Combination, in Patients With Advanced Solid/Metastatic Tumours.	I/II
NCT03842228	A Phase Ib Biomarker-Driven Combination Trial of Copanlisib, Olaparib, and Durvalumab (MEDI4736) in Patients With Advanced Solid Tumors	I

Assay Information

Methodology, Scope, and Application: The OCAv3 next-generation sequencing (NGS) assay identifies more than 3000 annotated mutations of interest (MOIs) broadly categorized into 5 mutation types: single nucleotide variants (SNVs), small insertions/deletions (Indels), large (>3 bases) insertions/deletions (Large Indels), copy number variants (CNVs), and gene fusions. The assay utilizes the Thermo Fisher Scientific Oncomine® Comprehensive Assay v3.0 (OCAv3), a next-generation sequencing (NGS) assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from formalin-fixed tissue for sequencing on the Ion S5 XL platform and analyzed by the current version of Torrent Suite Software and Ion Reporter. The OCAv3 NGS Assay currently can reliably identify the presence or absence of >3000 known mutations and polymorphisms in the 161 unique genes compared to the Human Reference Genome hg19, including the genes listed below. The OCAv3 assay is a laboratory developed test designed to find gene mutations within tumors (somatic mutations).

Analytical Sensitivity and Specificity: The OCAv3 NGS Assay has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 97.18% sensitivity compared with orthogonal assay results and 100% specificity for the 5 mutation types reported. 99.99% or greater reproducibility has been demonstrated. All quality measures for this assay were within defined assay parameters.

Hereditary and Germline Mutations: This assay examines tumor tissue only and does not examine normal (non-tumor) tissue. Mutations detected by the assay may be present only in the tumor, or in every cell of the body (including non-tumor cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or combination of features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's) it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-tumor) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent Oncomine Reporter software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of Oncomine Reporter. The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding a particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

AKT1, AKT2, AKT3, ALK, AR, ARAF, AXL, BRAF, BTK, CBL, CCND1, CDK4, CDK6, CHEK2, CSF1R, CTNNB1, DDR2, EGFR, ERBB2, ERBB3, ERBB4, ERCC2, ESR1, EZH2, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, FOXL2, GATA2, GNA11, GNAQ, GNAS, H3F3A, HIST1H3B, HNF1A, HRAS, IDH1, IDH2, JAK1, JAK2, JAK3, KDR, KIT, KNSTRN, KRAS, MAGOH, MAP2K1, MAP2K2, MAP2K4, MAPK1, MAX, MDM4, MED12, MET,

MTOR, MYC, MYCN, MYD88, NFE2L2, NRAS, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPP2R1A, PTPN11, RAC1, RAF1, RET, RHEB, RHOA, ROS1, SF3B1, SMAD4, SMO, SPOP, SRC, STAT3, TERT, TOP1, U2AF1, XPO1

Genes assayed with full exon coverage:

ARID1A, ATM, ATR, ATRX, BAP1, BRCA1, BRCA2, CDK12, CDKN1B, CDKN2A, CDKN2B, CHEK1, CREBBP, FANCA, FANCD2, FANCI, FBXW7, MLH1, MRE11, MSH2, MSH6, NBN, NF1, NF2, NOTCH1, NOTCH2, NOTCH3, PALB2, PIK3R1, PMS2, POLE, PTCH1, PTEN, RAD50, RAD51, RAD51B, RAD51C, RAD51D, RB1, RNF43, SETD2, SLX4, SMARCA4, SMARCB1, STK11, TP53, TSC1, TSC2

Genes assayed for copy number variations:

AKT1, AKT2, AKT3, ALK, AR, AXL, BRAF, CCND1, CCND2, CCND3, CCNE1, CDK2, CDK4, CDK6, EGFR, ERBB2, ESR1, FGF19, FGF3, FGFR1, FGFR2, FGFR3, FGFR4, FLT3, IGF1R, KIT, KRAS, MDM2, MDM4, MET, MYC, MYCL, MYCN, NTRK1, NTRK2, NTRK3, PDGFRA, PDGFRB, PIK3CA, PIK3CB, PPARG, RICTOR, TERT

Genes assayed for detection of fusions:

AKT2, ALK, AR, AXL, BRAF, BRCA1, BRCA2, CDKN2A, EGFR, ERBB2, ERBB4, ERG, ESR1, ETV1, ETV4, ETV5, FGFR1, FGFR2, FGFR3, FGR, FLT3, JAK2, KRAS, MDM4, MET, MYB, MYBL1, NF1, NOTCH1, NOTCH4, NRG1, NTRK1, NTRK2, NTRK3, NUTM1, PDGFRA, PDGFRB, PIK3CA, PPARG, PRKACA, PRKACB, PTEN, RAD51B, RAF1, RB1, RELA, RET, ROS1, RSP02, RSP03, TERT

DISCLAIMER

This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources, but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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